THE EVER-EXPANDING UNIVERSE — How Big, How Old, and Who Figured It Out (7.2)

Student: Ming (G7–G8, fast & advanced) | **Duration:** ~90 min | **Text:** Spectrum Science G7, Lesson 7.2, "The Ever-Expanding Universe" (pp. 130–131)

The narrative spine: The page is titled "The Ever-Expanding Universe" — and then never once mentions expansion. What it actually contains is the story of Earth losing its throne, one demotion at a time: geocentric (Earth at the center) → heliocentric (just a planet) → one star among many → one galaxy among billions. The real engine under the whole page: **nobody argued their way to a bigger universe. Instruments did.** Every time measurement got better, the universe got bigger — naked eye, then telescope, then photography, then better yardsticks. And the page stops at the exact moment it gets interesting: it names Hubble, says he "proved the existence of galaxies," and closes. This lesson keeps walking: Leavitt's yardstick → why models die → the redshift lab done with her own pencil → the expanding universe the title promised → and the discovery, in 1998, that the expansion is speeding up — so "ever-expanding" is a live question, not a settled one.

Teacher's thesis (don't state it until the paper lab's backwards trace): models don't die when someone has a better argument. They die when someone builds a better instrument. Ptolemy fit the data for 1,500 years — fit is not truth.

THE RUNNING THREAD — THE RED PEN (light edition). This page is not stale like 6.9 and not sloppy like 7.1 — its sins are different: one claim off by an order of magnitude, a title that promises a lesson the page never delivers, and a page that quietly contradicts its own sidebar. Three catches promised, two soft extras. Tally at the end.

Resources

- Textbook spread: screenshot in this folder — `textbook-pages-130-131.jpg` (book pages 130–131, title "The Ever-Expanding Universe").
- ☒ PhET, My Solar System — https://phet.colorado.edu/en/simulations/my-solar-system (HTML5, runs in a browser, no login; Alex screen-shares, Ming drives).
- ☒ **The Hubble Paper Lab** — worksheet in this folder — `hubble-paper-lab/hubble-paper-lab-worksheet.pdf` (page 1 = student, page 2 = teacher key; also as `worksheet-page-1.png` / `worksheet-page-2.png`).
- [VIDEO] SciShow Great Minds, "Henrietta Leavitt" — https://www.youtube.com/watch?v=2FrY6gRPC7k (transcript in this folder: `scishow-leavitt-transcript.txt`)
- [VIDEO] TED-Ed, "Three ways the universe could end" — https://www.youtube.com/watch?v=itpLU7OzNV8 (transcript in this folder: `teded-universe-endings-transcript.txt`)
- ☒ Stellarium (free planetarium, stellarium.org) — for the hook; Alex screen-shares, Ming drives.
- ☒ Galaxy Zoo — https://www.zooniverse.org/projects/zookeeper/galaxy-zoo
- ☒ Balloon + marker + pump — for Activity 7 (Alex has the pump).

Notes: the older `Concept Note — The Ever-Expanding Universe (7.2).md` in this folder is superseded by this plan. The retired custom build "The Cosmic Detective" was archived to `_archive/` on 2026-10-16 — it told the story interactively but not clearly (see Activity 5 for what replaced it).

ACTIVITY 1 — HOOK: HOW OLD IS THE LIGHT IN YOUR EYE? (7 min) • Stellarium, she drives

Format: Alex shares screen with Stellarium open. Ming controls the mouse. No agenda stated beyond the first question.

SAY THIS — the setup:

> "Point at that bright star — Sirius, the brightest in our sky. Click it, read me what it says. Distance: 8.6 light-years. That means the light hitting your eye right now left that star eight and a half years ago. You are not seeing Sirius. You are seeing its past. Now find Vega — 25 light-years. Polaris, the north star — over 400. Betelgeuse — over 500: that light started traveling when Columbus hadn't sailed yet. Deneb, in the Summer Triangle — a few thousand; that photon left when ancient Greece was young. Now zoom out and click the faint fuzzy patch in Andromeda — two and a half million light-years. That light is older than the human species."

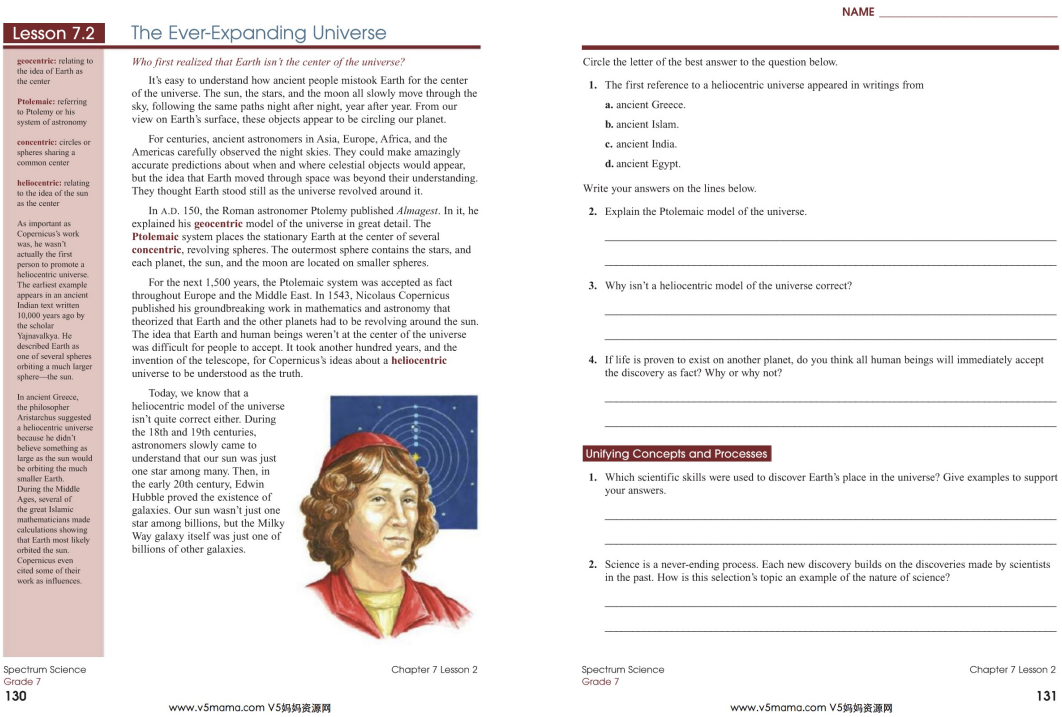
Let her drive for 3–4 minutes — click anything, read distances, find the oldest light she can. Let her discover a globular cluster or another galaxy if she stumbles on one.

SAY THIS — the mission:

> "So here's the thing nobody tells you in school: every telescope is a time machine, and the whole of astronomy is the art of reading old light. Today your textbook asks who first realized Earth isn't the center of the universe. But the deeper question hiding under it is — how did humans, stuck on one rock, ever figure out how big this place is? By the end of today you'll have done the measurements that changed the answer."

Why this hook: she operates the instrument herself (the lesson's thesis — instruments, not arguments — is enacted in the first seven minutes), and every distance she reads is a light-travel time, which quietly sets up Leavitt and Hubble without any math.

ACTIVITY 2 — TEXT + RED PEN (8 min) • reading



Lesson 7.2 The Ever-Expanding Universe

geocentric: relating to the idea of Earth as the center

Ptolemaic: referring to Ptolemy or his system of astronomy

concentric: circles or spheres sharing a common center

heliocentric: relating to the idea of the sun as the center

As important as Copernicus's work was, he wasn't actually the first person to promote a heliocentric universe. The earliest example appears in an ancient Indian text written 10,000 years ago by the scholar Yajnavalkya. He described Earth as one of several spheres orbiting a much larger sphere—the sun.

In ancient Greece, the philosopher Aristarchus suggested a heliocentric universe because he didn't believe something as large as the sun would be orbiting the much smaller Earth. During the Middle Ages, several of the great Islamic mathematicians made calculations showing that Earth most likely orbited the sun. Copernicus even cited some of their work as influences.

Who first realized that Earth isn't the center of the universe?

It's easy to understand how ancient people mistook Earth for the center of the universe. The sun, the stars, and the moon all slowly move through the sky, following the same paths night after night, year after year. From our view on Earth's surface, these objects appear to be circling our planet.

For centuries, ancient astronomers in Asia, Europe, Africa, and the Americas carefully observed the night skies. They could make amazingly accurate predictions about when and where celestial objects would appear, but the idea that Earth moved through space was beyond their understanding. They thought Earth stood still as the universe revolved around it.

In A.D. 150, the Roman astronomer Ptolemy published *Almagest*. In it, he explained his **geocentric** model of the universe in great detail. The **Ptolemaic** system places the stationary Earth at the center of several **concentric**, revolving spheres. The outermost sphere contains the stars, and each planet, the sun, and the moon are located on smaller spheres.

For the next 1,500 years, the Ptolemaic system was accepted as fact throughout Europe and the Middle East. In 1543, Nicolaus Copernicus published his groundbreaking work in mathematics and astronomy that theorized that Earth and the other planets had to be revolving around the sun. The idea that Earth and human beings weren't at the center of the universe was difficult for people to accept. It took another hundred years, and the invention of the telescope, for Copernicus's ideas about a **heliocentric** universe to be understood as the truth.

Today, we know that a heliocentric model of the universe isn't quite correct either. During the 18th and 19th centuries, astronomers slowly came to understand that our sun was just one star among many. Then, in the early 20th century, Edwin Hubble proved the existence of galaxies. Our sun wasn't just one star among billions, but the Milky Way galaxy itself was just one of billions of other galaxies.

Spectrum Science Grade 7 130 Chapter 7 Lesson 2

NAME _____

Circle the letter of the best answer to the question below.

1. The first reference to a heliocentric universe appeared in writings from
   a. ancient Greece.
   b. ancient Islam.
   c. ancient India.
   d. ancient Egypt.

Write your answers on the lines below.

2. Explain the Ptolemaic model of the universe.

3. Why isn't a heliocentric model of the universe correct?

4. If life is proven to exist on another planet, do you think all human beings will immediately accept the discovery as fact? Why or why not?

Unifying Concepts and Processes

1. Which scientific skills were used to discover Earth's place in the universe? Give examples to support your answers.

2. Science is a never-ending process. Each new discovery builds on the discoveries made by scientists in the past. How is this selection's topic an example of the nature of science?

Spectrum Science Grade 7 131 Chapter 7 Lesson 2

www.v5mama.com V5妈妈资源网 www.v5mama.com V5妈妈资源网

The textbook spread under audit — Spectrum Science G7, Lesson 7.2, pp. 130–131

Tracking task, given before she reads: "Two lists while you read. (a) Every demotion — each time the page moves Earth or the Sun or the Milky Way further from the center. In order. (b) You're a fact-checker: mark anything wrong, self-contradictory, or suspiciously precise. Three things worth marking. One of them is enormous — literally, a factor of ten."

For you: list (a) = Earth circles nothing special (Copernicus) → the Sun is one star among many (18th–19th c.) → the Milky Way is one galaxy among billions (Hubble).

The three catches:

1. **"The earliest example appears in an ancient Indian text written 10,000 years ago by the scholar Yajnavalkya."** Ten thousand years ago is the late Stone Age — before writing, before the wheel, before agriculture in most of the world. Yajnavalkya is real, and the text associated with him (the Shatapatha Brahmana) genuinely does describe the Earth as orbiting the sun — and it is dated by scholars to roughly **700–800 BCE**. So the direction of the claim (ancient India had heliocentric ideas long before Copernicus) is defensible. The number is off by an order of magnitude. Somebody padded a real fact with a zero — and the workbook's question 1 is built on top of this sand.
2. **The title gap.** This page is called "The Ever-Expanding Universe." Read the whole spread: Ptolemy, Copernicus, Hubble, galaxies — the word expansion appears exactly zero times. The title promises the punchline of the story and the page never tells it; worse, "ever-" (the expansion never stopping, even speeding up) is a 1998 discovery this book predates. The lesson's job today is to deliver what the title promised — Activity 6 does it with her own pencil.
3. **The page contradicts its own sidebar.** Main text: "the Ptolemaic system was accepted as fact throughout Europe and the Middle East." Sidebar, two centimeters away: "During the Middle Ages, several of the great Islamic mathematicians made calculations showing that Earth most likely orbited the sun. Copernicus even cited some of their work as influential." Both can't be quite true. What the main text means is "accepted as the working model by most astronomers" — which is different from "as fact," and different from "throughout the Middle East," where

astronomers were precisely the people improving on Ptolemy. The page flattens a millennium of real science into a shrug.

Soft extras, if she finds them: the workbook's question 1 spells it "helicentric" (missing the second c — a typo in a question about who spelled it first) • "It took another hundred years, and the invention of the telescope" — 1543 to 1608 is sixty-five; loose rounding • "Hubble proved the existence of galaxies" — what he actually proved, in 1924, was that Andromeda's Cepheid star sits outside the Milky Way; "galaxies" as a category was the consequence, not the experiment.

ACTIVITY 3 — WORKBOOK Q1–4 + UNIFYING CONCEPTS (10 min) • Q&A

Q1 (multiple choice). The first reference to a helicentric [sic] universe appeared in writings from⋯ → Per the page's own sidebar: ancient India. → Handle the shaky premise head-on (this is the Red Pen landing): the answer the page wants is India, and India is a respectable answer — Yajnavalkya's text really does put Earth in orbit, a couple of millennia before Copernicus. But the question's foundation is the 10,000-year claim, and that claim is wrong — so the MCQ is a castle on sand. This is what catch #1 was for. (And yes — "helicentric" is the book's typo, not ours.)

Q2. Explain the Ptolemaic model of the universe. → Expected: Earth stationary at the center, celestial bodies on concentric revolving spheres. → Enrich with the machinery the page skips: it never mentions **epicycles** — the little circles-upon-circles Ptolemy actually used to make the model fit the wandering planets. Sketch it for her on paper: a small circle whose center rides a bigger circle, a planet on the rim — that's the machine. Hold the sketch; Activity 5 shows what finally replaced it.

Q3. Why isn't a heliocentric model of the universe correct? → Expected: because the sun is just one star among many, in one galaxy among billions — the solar system has no claim on the center. → This is the page's quiet bombshell: it's not that the answer moved from Earth to the Sun. **There is no center.** Every frame they threw at the sky — Earth, Sun — was just another point. Activity 7 takes this question and makes her feel the answer.

Q4. If life is proven to exist on another planet, do you think all human beings will immediately accept the discovery as fact? → Expected: no — new ideas that demote humans meet resistance; Copernicus's model took roughly a century plus the telescope to win. → The pattern the page itself taught: acceptance tracks evidence you can see, not persuasion. Bridge, one line: "There's a whole lesson hiding in this question — aliens, SETI, how we'd even know. Someday." (Park it; that's a future class.)

UC1. Which scientific skills were used to discover Earth's place in the universe? → Systematic observation and record-keeping (centuries of sky logs) • model-building and fitting (Ptolemy) • instrumentation (the telescope) • mathematics (the Islamic astronomers' calculations) • classification and pattern-finding (Leavitt's Cepheid catalogs — coming in the next activity). Note out loud: every single one is a measurement skill.

UC2. Science is a never-ending process — how is this topic an example? → Each demotion looked like the final answer in its time; each was overturned by better data, not by better rhetoric. And the process didn't stop at the page's end — 1998 moved the goalposts again.

ACTIVITY 4 — [VIDEO] HENRIETTA LEAVITT: THE YARDSTICK (8 min) • video + questions

[VIDEO] https://www.youtube.com/watch?v=2FrY6gRPC7k (~4 min)

SAY THIS — before watching:

> "Your page says Hubble 'proved the existence of galaxies' in one sentence, like it was easy. It wasn't, and the reason it was even possible was a woman whose job title was 'computer' — whose job doesn't exist anymore because she and her colleagues were the computers. Watch for what she was allowed to touch, and what she figured out anyway."

Questions:

1. What was Leavitt's actual job, and why was she working for free? → A "computer" at Harvard Observatory — cataloguing stars on glass photographic plates. Women weren't allowed to direct research or even operate the telescopes; the observatory literally couldn't find men willing to do the tedious cataloguing, so it hired women.
2. Why was the Small Magellanic Cloud the perfect hunting ground? → It's one star cluster at (effectively) one distance — so all its stars are equally far away. Differences in how bright they look are differences in how bright they truly are. She could compare true brightnesses without knowing a single distance.
3. State her discovery — Leavitt's Law — in one sentence. → The longer a Cepheid variable's pulsing period, the brighter it truly is. Bigger star, slower heartbeat. The rhythm is the wattage label.
4. What was the "dim or far?" deadlock, and how did her law break it? → Before Leavitt, a faint-looking star was ambiguous — genuinely dim, or just far? Her law gave true brightness from the period; compare that to how bright it looks, and the difference is distance. Measuring range jumped from ~100 light-years to millions. One office of women who weren't allowed to touch the telescopes, and the universe doubled.
5. What did Hubble do with her law — and what couldn't she do with it? → He pointed it at Andromeda's Cepheids (1924) and got a distance beyond the Milky Way — the island universe proven; then combined distances with Slipher's redshifted spectra → expansion. Leavitt herself was never permitted telescope time to follow it up.
6. The Nobel story. → 1923: Swedish physicists wrote to Harvard asking for her work — she was being considered for the Physics Nobel. She had died of cancer three years earlier, at 53. Nobel Prizes are never awarded posthumously. Hubble himself said the prize was her due.

SAY THIS — the landing:

> "Notice what just happened: the single most important distance tool in astronomy was found by someone the system had filed under 'assistant.' In a few minutes you'll open a night assistant's log — the distances in it were measured with her law. Your half of the job is the light."

ACTIVITY 5 — ☒ PhET MY SOLAR SYSTEM: ONE RULE RUNS THE WHOLE SKY (10 min) • simulation

☒ https://phet.colorado.edu/en/simulations/my-solar-system

Why here: the page left Ptolemy vs. Copernicus unresolved — with eyes alone, both fit. This activity answers how the argument was actually settled: not by a better fit, but by a model that **generates the whole sky from one rule**. (This is the idea the retired Cosmic Detective's Act 1 was reaching for — done more honestly with a real, research-grade sim instead of a fake slider game.)

Steps (she drives, Alex narrates):

- **Round 1 — one planet.** Load the preset with a star and one planet. Press play. The orbit draws itself. → Q: "How many rules did I give it?" → **One: gravity.** Nothing was tuned; the ellipse falls out of the rule.
- **Round 2 — the whole sky.** Add a second planet farther out (or load the multi-planet setup) and run. Both orbit the same star, each in its own ellipse. → Q: "And how many circles did I have to hand-draw?" → Zero. Now hold up the epicycle sketch from Activity 3: Ptolemy needed a separate hand-tuned circle-machine for every planet, and his machine couldn't say why planets move at all — it only described.
- **Round 3 — interrogate the machine.** Grab the inner planet's velocity arrow and stretch it. The orbit stretches; push too hard and the planet leaves forever. That's what a model is: not a picture of the sky but a machine you can put questions to — and the sky itself won't let you do this experiment on demand.

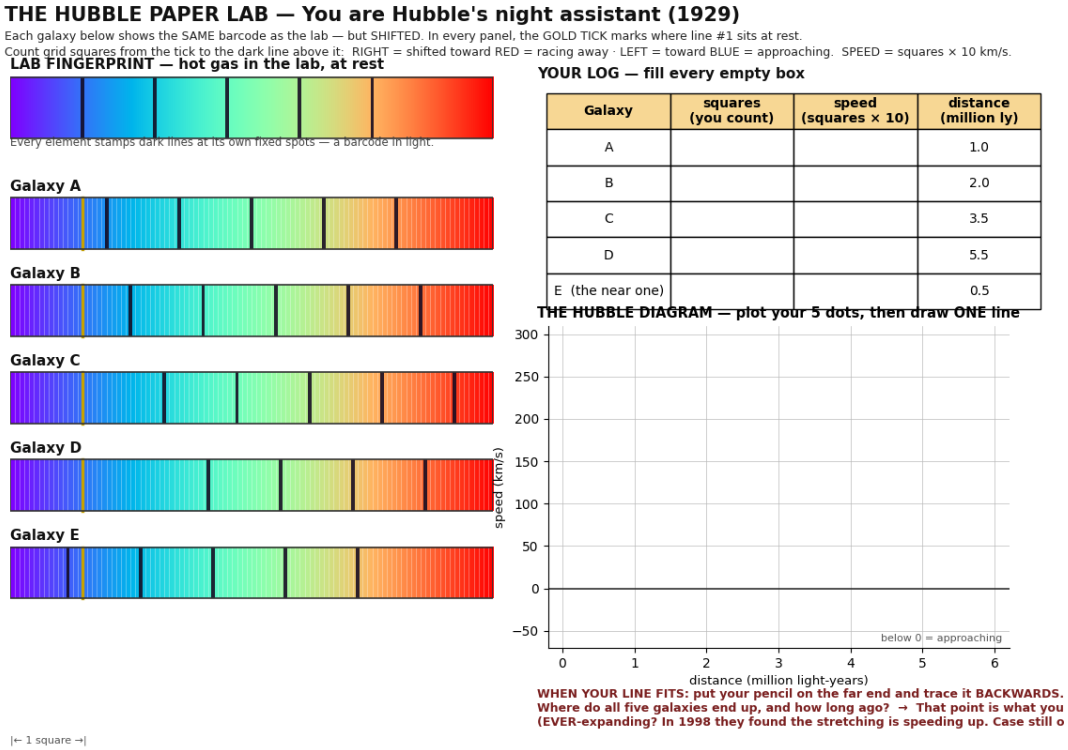
SAY THIS — the landing:

> "Ptolemy's machine had a fudge for every planet and explained nothing it wasn't told to fit. The gravity machine sets ONE rule and the whole clockwork runs — moons, comets, tides, and much later, why there's anything to argue about at all. So how did Copernicus actually win? Not the argument — the economy. More sky, fewer assumptions. That's how models die: not when they stop fitting, but when something else fits MORE with LESS."

Honest limits, for you: the sim shows orbits top-down. It does NOT show the sky from Earth's surface or retrograde loops — don't promise that. Its one job here: what a model is, and why 'it fits' wasn't enough to keep Ptolemy.

ACTIVITY 6 — ☒ THE HUBBLE PAPER LAB (12 min) • paper worksheet — the heart of the lesson

☒ `hubble-paper-lab/hubble-paper-lab-worksheet.pdf` — page 1 for Ming (print it or screen-share it), page 2 is your key. Also saved as PNGs. Everything on it is measured in grid squares, so it works on paper and over screen-share.



The Hubble Paper Lab student worksheet (print or screen-share; teacher key on worksheet page 2)

How to run it:

SAY THIS — the mission:

> "1929, Mount Wilson. You're Hubble's night assistant. The distance team — using Leavitt's law — already filled the last column of your log. Your job is the light. This strip is the lab fingerprint: hot gas at rest stamps dark lines at fixed spots — a barcode. Each galaxy shows the same barcode, shifted. Count the grid squares from the gold tick to the shifted line — right means red, racing away; left means blue, approaching. Speed is squares times ten. That's the only arithmetic in this lab."

- She counts squares for all five galaxies, fills the log (speeds: A 50 • B 100 • C 170 • D 260 km/s away; E 30 km/s **approaching** — E lands below the zero line on the diagram. If she asks why: the nearest galaxy wins the local tug-of-war — that's her Andromeda click from the hook).
- She plots her five dots and draws **one line** through the four receding dots and the corner. The corner matters: zero distance → zero speed. A line forced through that corner by the data is a law, not a coincidence — the farther, the faster.
- **RUN THE FILM BACKWARDS:** pencil on the far end of her line, trace it home. Where do all five galaxies end up, and how long ago? (~13.8 billion years.) That point is exactly what her page's title was hiding — Red Pen catch #2, redeemed by her own pencil.
- The 1998 button: "Ever-expanding? Astronomers found the stretching is speeding up. The case is still open." Leave it there — Activity 9 picks it up.

Teacher key (worksheet p.2): counts A=5, B=10, C=17, D=26 right; E=3 left → -30 km/s. Six talking points on the key page, including the honest credits: distances → Leavitt's law; shifts → mostly Slipher, before Hubble. Slope ≈ 50 km/s per million ly in this teaching model — the real value is ~22 per million ly; say "model" if asked.

Why paper and not a build: every number on this page is now hers — she counted the squares, she plotted the dots, she drew the line, she ran it backwards with a pencil. Nothing was revealed to her by a button. This is the experiment the page never did.

ACTIVITY 7 — ☒ THE NO-CENTER UNIVERSE (6 min) • balloon demo — fully scripted

☒ Balloon + marker + pump (you have the pump).

Bridge, say it: "Your workbook asked why heliocentric isn't correct, and the answer was there is no center. Your paper-lab line said every galaxy races away from every other. So — race away from where?"

- Inflate the balloon partway. Draw ~10 dots on it with the marker. Label one dot **US.**
- **Ask before inflating further:** "When I blow it up more, which dot does everything move away from?" Let her commit.
- Inflate. Every dot sees every other dot receding — pick any dot and it looks like the center of the expansion.
- The conclusion she should say out loud: **there is no special dot.**
- The kicker, every student's favorite: "So where did the Big Bang happen?" → Everywhere. The Big Bang wasn't an explosion in space — it was an event of space. Every dot (every galaxy) was there. "Where did it start?" quietly assumes space is a container the universe sits inside; space is the thing that started.
- Honesty note, say it if the balloon confuses: the balloon's surface is only a model of space — the 2D part of the rubber. Our universe is the 3D version of that surface. The inside and outside of the balloon are not "other universes" — they're nothing, they're where the model stops.

ACTIVITY 8 — ☒ GALAXY ZOO: CLASSIFY FOR REAL SCIENCE (7 min) • live citizen science

☒ https://www.zooniverse.org/projects/zookeeper/galaxy-zoo

Format: Alex shares screen; together they classify 2–3 galaxies. She calls every judgment; Alex clicks. Real project — her clicks join the catalog real astronomers use.

- Walk the decision tree out loud on galaxy #1: smooth or features? spiral arms? how many? any ring? central bulge? anything odd? There is no wrong answer — the science works because thousands of people vote.
- On galaxy #2: "Wait — we're looking at light that's been traveling for a hundred million years, deciding its shape from a thumbnail. Is that⋯ allowed?" → Yes, that's the whole job. (Instrument, again.)
- On galaxy #3, hunt a merger or a ring. If she gets one, that's the story of the day: two galaxies in mid-collision — expansion is about space stretching, not about galaxies politely staying apart (balloon callback).

ACTIVITY 9 — [VIDEO] THREE WAYS THE UNIVERSE COULD END (8 min) • video + questions

[VIDEO] https://www.youtube.com/watch?v=itpLU7OzNV8 (~5 min)

SAY THIS — before watching:

> "Your page is called 'The Ever-Expanding Universe' and never asks the obvious follow-up: expanding — forever? The paper lab left the case open. The universe has a past; it apparently also has a future. Three candidates. Place your bet before we watch."

Questions:

1. The video's whole model is two objects, a rubber band, and two hooks. What are they? → The objects are galaxies; the rubber band is gravity pulling them together; the hooks are the repulsive expansion (dark energy's push). The end of the universe is the outcome of that tug-of-war.
2. Walk the Big Freeze. What happens, in order? → Galaxy clusters separate → galaxies dissolve into lonely objects → star-forming gas runs out, stars fade → light redshifts to faint long wavelengths → dark and cold forever. Also called the Big Chill or Heat Death. Current observations point here.
3. The Big Rip — what does it tear, and in what order? → First galaxies, then solar systems, and finally the very forces — electromagnetic, weak, strong — that hold atoms and nuclei together. Matter itself comes apart. (She's met the strong force — Fission vs Fusion callback.)
4. The Big Crunch, then the Big Bounce. What does "each bounce wipes away any record of the universe's previous history" mean for science? → If true, no instrument, ever, could count how many bounces came before us — the information is destroyed. A universe where the deepest question is permanently unanswerable.
5. What does the answer depend on — and how long do we have in the leading scenario? → The shape of the universe, the amount of dark energy, and how the expansion rate changes. Big Freeze: about 10^100 years before the chill sets in — don't stock up on mittens.

SAY THIS — the landing:

> "Notice something: the video's rubber band and hooks — that's your balloon, one outfit later. When two different activities keep handing you the same toy, it's because nature only has so many."

ACTIVITY 10 — RED PEN TALLY + 3-2-1 (5 min) • reflection

Step 1 — count the catches. Three promised, two soft (the "helicentric" typo and the "hundred years" rounding — plus "Hubble proved the existence of galaxies" if she grabbed it). Character of this page, say it: "6.9 was stale, 7.1 was careless — this one overpromised. The title told you the answer before the page asked the question, and one sidebar claim was inflated by a factor of ten. Different disease each time. That's why you fact-check everything, forever. And today you did the experiment the title promised."

Step 2 — 3-2-1:

- 3 things this page was hiding (expect: the title's missing expansion · the Islamic astronomers the main text erased · the epicycles the model actually ran on · the woman with the yardstick).
- 2 connections to previous lessons (expect: the Parallax lesson — another rung of the same distance ladder · Fission vs Fusion — the strong force in the Big Rip · the Signal Chain — reading a signal out of noise, redshift edition · The Gap — what it costs when a measurement can't be made · Sports Science — no, wrong Ming; try guitar/pickup and Faraday from the music lessons if she reaches).
- 1 question that remains (expect: what is dark energy · what was before the first day · will it really expand forever).

[BONUS] BONUS ACTIVITIES — if you're ahead of schedule

B1 — The 1054 guest star: China's supernova (5 min) · storytelling The page mentions ancient astronomers in Asia in one sentence and moves on. China's court astronomers were the best record-keepers on Earth: in July 1054 (Song dynasty) they recorded a "guest star" in Taurus so bright it was visible in daylight for 23 days, and at night for nearly two years. That star's corpse is still there — the Crab Nebula — and today astronomers can date the explosion to days using those Chinese records. European sky-watchers, by contrast, wrote almost nothing — some records were likely suppressed for superstitious reasons. Q: "Why do Chinese court records matter to a modern astronomer with a space telescope?" → Because instruments measure the present; old, careful notebooks are the only time machines we have for the past. (Full-circle callback to the hook: telescopes are time machines — and so are logbooks.)

B2 — Scale of the Universe 2 (4 min) · guided tour https://htwins.net/scale2/ — the famous zoom from a coffee bean (10^0 m) down to a Planck length and up to the observable universe (10^27 m). Drive it slowly, narrating: every factor of ten is one keypress — that's the log scale she knows from the pH lesson, now as space. Stop at: a human cell, DNA, Earth, the Sun, Betelgeuse (the hook's old-light star — now she sees its size), the Milky Way, the Local Group. Q: "Where on this zoom does the word 'center' stop meaning anything?" → Somewhere around the Sun: from Earth outward, every frame she was taught was a center just becomes another pixel.

B3 — The expanding-ruler thought experiment (3 min) · pure discussion, no props An ant lives on a one-meter ruler. You stretch the ruler to two meters. The ant's steps are the same size — but its home is now farther from everything. Question: from inside the ruler, how would the ant ever discover the stretching? → Everything it measures with (its legs, light, atoms) stays the same size; only the gaps grow. That's why cosmic expansion only shows up across galaxy-sized gaps — atoms, people, and solar systems don't expand: their own forces (the rubber band) beat the stretching locally. This is also Galaxy E's excuse from the paper lab. Two minutes, one big idea.

Summary & Flow

#	Activity	Type	Min
1	How Old Is the Light in Your Eye? — hook	Stellarium, she drives	7

#	Activity	Type	Min
2	Text + Red Pen (3 catches)	Reading	8
3	Workbook Q1–4 + Unifying Concepts	Q&A	10
4	[VIDEO] Henrietta Leavitt + 6 Qs	Video + Q&A	8
5	☒ PhET My Solar System — one rule, whole sky	Simulation	10
6	☒ The Hubble Paper Lab — her pencil draws the line	Paper worksheet	12
7	☒ The No-Center Universe — balloon	Demo	6
8	☒ Galaxy Zoo — classify 2–3 together	Live citizen science	7
9	[VIDEO] Three Ways the Universe Could End + 5 Qs	Video + Q&A	8
10	Red Pen tally + 3-2-1	Reflection	5
—	[BONUS] B1 The 1054 Guest Star — China's supernova	Storytelling	5
—	[BONUS] B2 Scale of the Universe 2	Guided interactive	4
—	[BONUS] B3 The Expanding Ruler	Discussion	3

Core total ≈ 81 min — trim one of: Activity 4 to 6 min (cut Q5/Q6, tell the Nobel story in one line yourself), Activity 8 to 5 min (two galaxies), Activity 9 to 6 min (cut Q4). Text + workbook ≈ 18 min ≈ 20% ✓ • 10 core activities ✓ • alternation holds (sim → paper → demo → live data → video, no two adjacent of the same kind) ✓ • 2 videos + 1 PhET + 1 paper lab + 1 live citizen-science project + 1 planetarium + 1 physical demo ✓ • 4 read-aloud blocks ✓ • math at exactly one operation (squares × 10), stated out loud ✓

Cut list, in order: Activity 4 to 6 min (Q5/Q6 folded into one line) • Activity 8 to 5 min • Activity 9 to 6 min. Do not cut: the hook (she drives) • catch #2 (the title gap — it's the lesson's premise) • the paper lab's backwards trace (that IS the lesson) • the balloon's "where did it start?" exchange. Those four are the lesson.

Bookends: the hook makes her read light-travel time off the sky with her own clicks → Leavitt gives the yardstick and the video gives her the human → PhET shows what kills a model (more sky, fewer assumptions) → the paper lab puts the expansion law in her own hand, drawn with her own pencil → the balloon makes "no center" physical → Galaxy Zoo shows the real, messy sky → the endings video pays off the title's "ever-" → the tally closes the Red Pen. One continuous question the whole class: how do you know, when you can't go there? You measure old light.

(!) FACT-CHECK BEFORE CLASS

The load-bearing claims, worth ten minutes:

- Star light-travel distances for the hook (approximate is fine — say "roughly"): Sirius 8.6 ly ✓ • Vega ~25 ly ✓ • Polaris ~430 ly ✓ • Betelgeuse ~550–650 ly (estimates vary) ✓ • Deneb ~1,400–2,600 ly (very uncertain — say "a few thousand") • Andromeda ~2.5 million ly ✓.
- Yajnavalkya / Shatapatha Brahmana: dated ~700–800 BCE (some scholars say ~900 BCE). The heliocentric-ish passage is real; the "10,000 years" is wrong by an order of magnitude. Don't

overclaim in the other direction — the catch is the number, not the idea. ✓

- Aristarchus of Samos: 3rd century BCE heliocentric proposal ✓ (sidebar gets this right). Ptolemy's Almagest ~150 CE ✓. Copernicus, De revolutionibus, 1543 ✓. First telescopes 1608 ✓ — so "another hundred years" is really ~65.
- Leavitt: Harvard "computer," Small Magellanic Cloud Cepheids, period–luminosity relation published 1912 (1908 paper had the early hints) ✓. Pre-Leavitt distance reach ~100 ly ✓. Hubble's 1924 Andromeda Cepheid → "island universe" ✓; 1929 velocity–distance relation (using Slipher's redshifts, mostly measured 1912–1925; Milton Humason's follow-up spectra) — if she asks "did Hubble discover redshift?", the honest answer is no: Slipher measured the shifts; Hubble measured the distances and put them together.
- "Hubble proved the existence of galaxies": softened — he proved Andromeda lies beyond the Milky Way; present as "loose wording," not "error."
- 1998: two teams (High-z Supernova Search, Supernova Cosmology Project — Riess, Perlmutter, Schmidt) find acceleration ✓; Nobel 2011 ✓. Cosmic microwave background afterglow: Penzias & Wilson, 1965 ✓.
- Universe age ~13.8 billion years (the TED-Ed transcript rounds to 14 — fine). Heat Death ~10^100 years is a "don't stock up on mittens" quote from the video — keep it attributed.
- The paper lab's values are teaching models: slope ≈ 50 km/s per Mly on the worksheet vs. the real Hubble constant ≈ 70 km/s per Mpc ≈ 21–22 per Mly; galaxy distances/speeds are illustrative. Same disclaimer as ever: "good enough to think with, not to navigate by."
- Andromeda is genuinely blueshifted (approaching ~110 km/s — local gravity wins) ✓. Galaxy Zoo: real, ongoing Zooniverse project; volunteer classifications feed published research ✓. PhET My Solar System is an HTML5 sim — runs in any modern browser, no Java ✓.

Two places where being imprecise is the point: Deneb's distance (astronomers genuinely disagree by a factor of two — model how to hold uncertainty) and "where did the Big Bang happen?" (the question itself is the lesson — every answer that sounds satisfying smuggles in an outside).